



The Sun — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Chapter 01_iii_The_Sun

Topic Index

1. Sun ki Composition aur Energy Source
2. Solar Eclipse
3. Halo aur Optical Phenomena
4. Astronomical Unit aur Earth-Sun Distance
5. Earth ki Motion — Rotation vs Revolution

Sun ki Composition aur Energy Source

- Sun ke centre mein matter **gas aur plasma** state mein hota hai (ionised gas). [Q1]
- Solar energy **nuclear fusion** se produce hoti hai (nuclear fission se NAHI) — light nuclei combine hoke heavier nucleus banate hain. [Q2]
- **Photosphere** = Sun ki bright outer/visible layer, temperature $\sim 5,500\text{--}6,000^\circ\text{C}$. [Q15]

Solar Eclipse

- Solar eclipse tab hota hai jab **Moon Earth aur Sun ke beech** aa jaata hai — sirf **New Moon/Amavasya** ke din possible hai. [Q4, Q5, Q6]
- Total solar eclipse sirf limited geographical region mein dikhta hai kyunki Moon ki shadow Earth ke cross-section se chhoti hoti hai. [Q7]
- **Diamond Ring effect** totality ke beginning AUR end dono par dikhta hai. [Q3]



EXAM TRAP

- Solar eclipse **New Moon/Amavasya** pe hota hai, Lunar eclipse **Full Moon/Poornima** pe — dono ko mix mat karo. [Q4, Q5, Q6]

Halo aur Optical Phenomena

- Sun ka **Halo** — cirrus clouds mein ice crystals se light refraction ke through banta hai. [Q8]

Astronomical Unit aur Earth-Sun Distance

- **Astronomical Unit (AU)** = Earth aur Sun ke beech average distance = ~ 149.6 million km (149.598×10^6 km). [Q9, Q10, Q11]
- Earth Sun ke sabse close **3 January** ko hoti hai — isko **Perihelion** kehte hain; farthest ~ 4 July ko — **Aphelion**. [Q12]
- Sun ka diameter Earth se ~ 109 times zyada hai. [Q13]

Earth ki Motion — Rotation vs Revolution

- Statement check: 'Rotation west-to-east se seasonal changes hote hain' — **GALAT** (rotation se day-night hota hai); 'Revolution se day-night' bhi **GALAT** (revolution se seasons hote hain) — dono statements incorrect hain. [Q14]

EXAM TRAP

- **Rotation** (axis pe ghumna) = Day-Night; **Revolution** (Sun ke around) = Seasons — is swap ko exam mein test kiya jaata hai. [Q14]